Washing program ROM listing

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2

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114 used words; ROM locations 72-FF are zero-filled NOPs. program.hex contains all 256 words in Logisim v2.0 raw format. Addresses and words below are hexadecimal. Mode parameters are decimal in assembly. The complete commented source is program.asm.

Address	Hex	Assembly
00	1E03	LDI R7, 3
01	1A02	LDI R5, 2
02	1CF0	LDI R6, 0xF0
03	6180	LD R0, [R6]
04	51C0	AND R0, R7
05	41C0	SUB R0, R7
06	A002	BNEZ R0, WAIT_START
07	1CFA	LDI R6, 0xFA
08	1001	LDI R0, 1
09	7180	ST R0, [R6]
0A	1CFC	LDI R6, 0xFC
0B	6180	LD R0, [R6]
0C	1201	LDI R1, 1
0D	5040	AND R0, R1
0E	9002	BEQZ R0, WAIT_START
0F	1CFB	LDI R6, 0xFB
10	6180	LD R0, [R6]
11	A016	BNEZ R0, MODE_HIGH
12	1464	LDI R2, 100
13	1650	LDI R3, 80
14	1828	LDI R4, 40
15	B019	JMP SETUP
16	1496	LDI R2, 150
17	1678	LDI R3, 120
18	183C	LDI R4, 60
19	1CF4	LDI R6, 0xF4
1A	1021	LDI R0, 0x21
1B	7180	ST R0, [R6]
1C	1CF5	LDI R6, 0xF5
1D	1001	LDI R0, 1
1E	7180	ST R0, [R6]
1F	1CF1	LDI R6, 0xF1
20	6180	LD R0, [R6]
21	D080	CMPGEU R0, R2
22	901F	BEQZ R0, FILL
23	1CF4	LDI R6, 0xF4
24	1020	LDI R0, 0x20
25	7180	ST R0, [R6]
26	1CF7	LDI R6, 0xF7
27	7780	ST R3, [R6]
28	1CF8	LDI R6, 0xF8
29	1003	LDI R0, 3
2A	7180	ST R0, [R6]
2B	1224	LDI R1, 0x24

Address	Hex	Assembly
2C	1CF4	LDI R6, 0xF4
2D	7380	ST R1, [R6]
2E	1CFC	LDI R6, 0xFC
2F	6180	LD R0, [R6]
30	5140	AND R0, R5
31	A069	BNEZ R0, ABORT
32	1CF9	LDI R6, 0xF9
33	6180	LD R0, [R6]
34	A03C	BNEZ R0, DRAIN_START
35	1024	LDI R0, 0x24
36	4040	SUB R0, R1
37	903A	BEQZ R0, REVERSE
38	1224	LDI R1, 0x24
39	B02C	JMP WASH_DIR
3A	122C	LDI R1, 0x2C
3B	B02C	JMP WASH_DIR
3C	1CF4	LDI R6, 0xF4
3D	1030	LDI R0, 0x30
3E	7180	ST R0, [R6]
3F	1CF8	LDI R6, 0xF8
40	1000	LDI R0, 0
41	7180	ST R0, [R6]
42	1CF5	LDI R6, 0xF5
43	1002	LDI R0, 2
44	7180	ST R0, [R6]
45	1CF1	LDI R6, 0xF1
46	6180	LD R0, [R6]
47	120A	LDI R1, 10
48	D200	CMPGEU R1, R0
49	9245	BEQZ R1, DRAIN
4A	1CF4	LDI R6, 0xF4
4B	1020	LDI R0, 0x20
4C	7180	ST R0, [R6]
4D	1CF7	LDI R6, 0xF7
4E	7980	ST R4, [R6]
4F	1CF8	LDI R6, 0xF8
50	1003	LDI R0, 3
51	7180	ST R0, [R6]
52	1CF4	LDI R6, 0xF4
53	1024	LDI R0, 0x24
54	7180	ST R0, [R6]
55	1CFC	LDI R6, 0xFC
56	6180	LD R0, [R6]
57	5140	AND R0, R5
58	A069	BNEZ R0, ABORT
59	1CF9	LDI R6, 0xF9
5A	6180	LD R0, [R6]
5B	9055	BEQZ R0, SPIN
5C	1CF4	LDI R6, 0xF4
5D	1020	LDI R0, 0x20
5E	7180	ST R0, [R6]
5F	1CF8	LDI R6, 0xF8
60	1000	LDI R0, 0

Address	**Hex**	**Assembly**
61	7180	ST R0, [R6]
62	1CF5	LDI R6, 0xF5
63	10AA	LDI R0, 0xAA
64	7180	ST R0, [R6]
65	1CF4	LDI R6, 0xF4
66	1040	LDI R0, 0x40
67	7180	ST R0, [R6]
68	C000	HALT
69	1CF4	LDI R6, 0xF4
6A	1000	LDI R0, 0
6B	7180	ST R0, [R6]
6C	1CF8	LDI R6, 0xF8
6D	7180	ST R0, [R6]
6E	1CF5	LDI R6, 0xF5
6F	10EE	LDI R0, 0xEE
70	7180	ST R0, [R6]
71	B002	JMP WAIT_START